DEDUCTION AND STATIC EQUIVALENCE ARE DECIDABLE FOR SEPARATE VARIABLE-PERMUTING THEORIES

SERDAR ERBATUR, ANDREW M. MARSHALL, PALIATH NARENDRAN, CHRISTOPHE RINGEISSEN

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Allowed standard prerequisites (editorial).

- Term rewriting, unification, frames, static equivalence and standard source undecidability results. Complete author §2 definitions. Named results on optimally reducing unification and the exact PCP rewrite system remain cited as author references; new frame/cut and contracting arguments retained.

2 Preliminaries

We use the standard notation of equational unification [7] and term rewriting systems [6]. Given a first-order signature Σ and a (countable) set of variables V, the Σ-terms over variables V are built in the usual way by taking into account the arity of each function symbol in Σ. Each Σ-term is well-formed: if it is rooted by a n-ary function symbol in Σ, then it has necessarily n direct subterms. For any term t, $|t|$ denotes the number of symbols occurring in t. The set of Σ-terms over variables V is denoted by $T(\Sigma, V)$. The set of variables from V occurring in a term $t \in T(\Sigma, V)$ is denoted by $Var(t)$. A term t is ground if $Var(t) = \emptyset$. A term t is linear if each variable in $Var(t)$ occurs only once in t. For any position p in a term t (including the root position ε), $t(p)$ is the symbol at position p, $t|_p$ is the subterm of t at position p, and $t[u]_p$ is the term t in which $t|_p$ is replaced by u. A substitution is an endomorphism of $T(\Sigma, V)$ with only finitely many variables not mapped to themselves. A substitution is denoted by $\sigma = \{x_1 \mapsto t_1, \ldots, x_m \mapsto t_m\}$, where the domain of σ is $Dom(\sigma) = \{x_1, \ldots, x_m\}$ and the range of σ is $Ran(\sigma) = \{t_1, \ldots, t_m\}$. Application of a substitution σ to t is written $t\sigma$. A Σ-equation is a pair of Σ-terms denoted by $s =^? t$ or simply $s = t$ when it is clear from the context that we do not refer to an axiom.

Let V_C denote a finite set of context variables (sometimes called holes) such that $V_C \cap V = \emptyset$. We use the notation $\diamond_i$ to represent the context variable $\diamond_i \in V_C$. A context is a linear term containing only context variables. More formally, a context is a term, $C \in T(\Sigma, V_C)$, where each variable (context hole) occurs at most once. Thus, the size of a context follows from the size of a term, where any hole occurrence counts for 1. Given a context C with m variables and m terms $S_1, \ldots, S_m \in T(\Sigma, V)$, $C[S_1, \ldots, S_m]$ denotes the term $C\{\diamond_1 \mapsto S_1, \ldots, \diamond_m \mapsto S_m\}$ where for $i = 1, \ldots, m$, $\diamond_i$ denotes the i-th context variable occurring in the term C, and C is called the context part of $C[S_1, \ldots, S_m]$. When $S_1, \ldots, S_m \in \mathcal{S}$, $C[S_1, \ldots, S_m]$ is said to be a context instantiated with terms in $\mathcal{S}$.

2.1 Equational Theories

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Given a set E of Σ-axioms (i.e., pairs of terms in $T(\Sigma, V)$, denoted by $l = r$), the equational theory $=_E$ is the congruence closure of E under the law of substitutivity (by a slight abuse of terminology, E is often called an equational theory). Equivalently, $=_E$ can be defined as the reflexive transitive closure $\leftrightarrow_E^*$ of an equational step $\leftrightarrow_E$ defined as follows: $s \leftrightarrow_E t$ if there exist a position p of s, $l = r$ (or $r = l$) in E, and substitution σ such that $s|_p = l\sigma$ and $t = s[r\sigma]_p$. An axiom $l = r$ is regular if $Var(l) = Var(r)$. An axiom $l = r$ is collapse-free if l and r are non-variable terms. An equational theory is regular (resp., collapse-free) if all its axioms are regular (resp., collapse-free). An equational theory E is said to be permutative if for any $l = r$ in E, the number of occurrences of any (function or variable) symbol in l is equal to the number of occurrences of that symbol in r. Well-known theories such as Associativity ($A = \{(x + y) + z = x + (y + z)\}$), Commutativity ($C = \{x + y = y + x\}$), and Associativity-Commutativity ($AC = A \cup C$) are permutative theories. A theory E is said to be shallow if variables can only occur at a depth at most 1 in axioms of E. For example, C is shallow but A and AC are not. Any constant in Σ not occurring in E is said to be free.

Definition 1 (Unification).
Given a signature Σ, an equational theory E, and a set of Σ-equations, $\Gamma = \{s_1 =^? t_1, \ldots, s_n =^? t_n\}$. The E-unification decision problem asks if there exists a substitution σ such that $s_i\sigma =_E t_i\sigma$ for all $1 \leq i \leq n$. If E is presented as a rewrite relation R, then we ask if there exists a substitution σ such that $s_i\sigma \leftrightarrow_R^* t_i\sigma$ for all $1 \leq i \leq n$.

Another important algorithmic problem is matching, which can be seen as a restricted form of the unification problem.

Definition 2 (Matching).
Given a signature Σ, an equational theory E, and a set of Σ-equations, $\Gamma = \{s_1 =^? t_1, \ldots, s_n =^? t_n\}$. The E-matching decision problem asks if there exists a substitution σ such that $s_i\sigma =_E t_i$ for all $1 \leq i \leq n$. If E is presented as a rewrite relation R, then we ask if there exists a substitution σ such that $s_i\sigma \leftrightarrow_R^* t_i$ for all $1 \leq i \leq n$.

2.2 Rewrite Relations

Given a signature Σ, an oriented Σ-axiom is called a rewrite rule of the form $l \to r$ such that $l, r \in T(\Sigma, V)$, l is not a variable and $Var(r) \subseteq Var(l)$. A finite set of rewrite rules is called a term rewriting system (TRS, for short). Let R be any TRS. For any Σ-terms s and t, s R-rewrites to t, denoted by $s \to_R t$, if there exist a position p of s, $l \to r \in R$, and substitution σ such that $s_{|p} = l\sigma$ and $t = s[r\sigma]_p$. The term s is said to be R-reducible, $s_{|p}$ is called a redex, and in the particular case where $s_{|p} = l\sigma$, s R-rewrites to t, denoted by $s \to_R t$. A term is an innermost redex if none of its proper subterms is a redex. The symmetric relation $\leftarrow_R \cup \to_R$ is denoted by $\longleftrightarrow_R$. The rewrite relation $\to_R$ is confluent if $\longleftrightarrow_R^*$ is included in $\to_R^* \circ \leftarrow_R^*$. The rewrite relation $\to_R$ is convergent if $\to_R$ is both terminating and confluent. When $\to_R$ is convergent, we have that for any terms t, t', $t \longleftrightarrow_R^* t'$ iff $t{\downarrow_R} = t'{\downarrow_R}$, where $t{\downarrow_R}$ (resp., $t'{\downarrow_R}$) denotes the unique normal form of t (resp., t') w.r.t $\to_R$. A TRS R is said to be optimally reducing if for any $l \to r$ in R and any substitution θ, the following holds: ($t\theta$ is R-irreducible for any strict subterm t of l) implies that $r\theta$ is R-irreducible. A TRS is linear if for any $l \to r$ in R, both l and r are linear. A convergent TRS R is said to be subterm convergent if for any $l \to r \in R$, r is either a strict subterm of l or a constant. We refer to [6] for the classical notion of overlap and critical pair. Hence, a trivial critical pair is obtained by an overlap at the root position of a rule with a renaming of the same rule. A TRS with no non-trivial critical pairs is non-overlapping.

2.3 Knowledge Problems

The applied pi calculus and frames are used to model attacker knowledge [3]. In this model, the set of messages or terms which the attacker knows, and which could have been obtained from observing one or more protocol sessions, are the set of terms in $Ran(\sigma)$ of the frame $\phi = \nu\tilde{n}.\sigma$, where σ is a substitution ranging over ground terms. We also need to model cryptographic concepts such as nonces, keys, and publicly known values. We do this by using names, which are essentially free constants. Here also, we need to track the names which the attacker knows, such as public values, and the names which the attacker does not know a priori, such as freshly generated nonces. $\tilde{n}$ consists of a finite set of restricted names; these names represent freshly generated names which remain secret from the attacker. The set of names occurring in a term t is denoted by $fn(t)$. For any frame $\phi = \nu\tilde{n}.\sigma$, let $fn(\phi)$ be the set of names $fn(\sigma)\backslash\tilde{n}$ where $fn(\sigma) = \bigcup_{t \in Ran(\sigma)} fn(t)$; and for any term t, let $t\phi$ denote – by a slight abuse of notation – the term $t\sigma$. For any term t, we say that t satisfies the name restriction of ϕ if $fn(t) \cap \tilde{n} = \emptyset$.

Definition 3 (Deduction).

Let $\phi = \nu\tilde{n}.\sigma$ be a frame, and t a ground term. We say that t is deduced from ϕ modulo E, denoted by $\phi \vdash_E t$, if there exists a term ζ such that $\zeta\sigma =_E t$ and $fn(\zeta) \cap \tilde{n} = \emptyset$. The term ζ is called a recipe of t in ϕ modulo E.

Another form of knowledge is the ability to tell if two frames are statically equivalent modulo E.

Definition 4 (Static Equivalence).

Two terms s and t are equal in a frame $\phi = \nu\tilde{n}.\sigma$ modulo an equational theory E, denoted $(s =_E t)\phi$, if $s\sigma =_E t\sigma$, and $\tilde{n} \cap (fn(s) \cup fn(t)) = \emptyset$. The set of all equalities $s = t$ such that $(s =_E t)\phi$ is denoted by $Eq(\phi)$. Given a set of equalities Eq, the fact that $(s =_E t)\phi$ for all $s = t \in Eq$ is denoted by $\phi \models Eq$. Two frames $\phi = \nu\tilde{n}.\sigma$ and $\psi = \nu\tilde{n}.\tau$ are statically equivalent modulo E, denoted as $\phi \approx_E \psi$, if $Dom(\sigma) = Dom(\tau)$, $\phi \models Eq(\psi)$ and $\psi \models Eq(\phi)$.

Deduction and static equivalence are known to be decidable for subterm convergent rewrite systems [2].

2.4 Graph Embedded TRS

It will be useful in one of the dichotomy results that follows to use a restricted class of term rewrite systems for which it is already known that the static equivalence problem is decidable. These systems were introduced in [25] and then simplified in [10], where they are used to expand on the class of subterm convergent term rewrite systems. Actually, we don't need the full class here for the proofs that follow. Therefore, we only introduce the needed components below, which keeps the definition simpler and more compact. In particular, we introduce a subset of contracting TRSs below that omit the permutative component. One can find the full definition in [10].

The class of contracting TRS first starts by modeling the graph minor relation in the TRS setting. This is done by a set of rewrite schemas. This set of rewrite schemata then induces a graph-embedded term rewrite system. Notice that this is very similar to what is often done when considering the homeomorphic embedding relation in TRSs.

Definition 5 (Graph Embedding).

Consider the following reduction relation, $\rightarrow^*_{R_{gemb}}$, where R_{gemb} is the set of rules given by the instantiation of the following rule schema:

$$\left\{\begin{array}{ll} & \text{for any } f \in \Sigma \\ (1) & f(x_1,\ldots,x_n) \to x_i \\ (2) & f(x_1,\ldots,\ x_{i-1},\ x_i,\ x_{i+1}\ \ldots,\ x_n) \to f(x_1,\ldots,\ x_{i-1},\ x_{i+1},\ \ldots,\ x_n) \\ & \text{and for any } f,g \in \Sigma \\ (3) & f(x_1,\ldots,x_{i-1},g(\bar{z}),x_{i+1},\ \ldots,x_m) \to g(x_1,\ldots,\ x_{i-1},\bar{z},x_{i+1},\ \ldots,x_m) \\ (4) & f(x_1,\ldots,x_{i-1},g(\bar{z}),x_{i+1},\ \ldots,x_m) \to f(x_1,\ldots,\ x_{i-1},\bar{z},x_{i+1},\ \ldots,x_m) \end{array}\right\}$$

We say a term t' is graph-embedded in a term t, denoted $t \succcurlyeq_{gemb} t'$, if t' is a well formed term such that $t \to^*_{R_{gemb}} t'$.

A TRS R is graph-embedded if for any $l \to r \in R$, $l \succcurlyeq_{gemb} r$ or r is a constant.

The graph embedded class is not sufficient to obtain decidability of static equivalence, so we next need to introduce a type of projection rule that ensures access to subterms and thus decidability of static equivalence.

Definition 6 (Projecting Rule).

Let R be a TRS, x a variable and t a term with a single occurrence of x. A projecting rule in R over t leading to x is a rule in R which is a variant of $t' \to x$ where t' is a superterm of t with no additional occurrences of x.

How these projecting rules relate to the entire term-rewrite system can now be defined.

Definition 7 (Projection-Closed Derivation).

Let R be a TRS. A non-empty R_{gemb}-derivation is said to be projection-closed with respect to R if it has the form

$$s \to^{l\to r,\gamma}_{R_{gemb}} s' \to^*_{R_{gemb}} t$$

where

- s is a linear term and t is a well formed term,
- $l \to r$ is a R_{gemb} rule among (1), (2) and (4),
- γ is a substitution ranging over variables,
- if $l \to r$ is rule (1), $f(x_1,\ldots,x_n) \to x_i$, then for any $x_i\gamma$ occurring in t there exists a projecting rule in R over $f(x_1,\ldots,x_n)\gamma$ leading to $x_i\gamma$,
- if $l \to r$ is rule (4), $f(x_1,\ldots,x_{i-1},g(\bar{z}),x_{i+1},\ldots,x_m) \to f(x_1,\ldots,x_{i-1},\bar{z},x_{i+1},\ldots,x_m)$, then for any $z_i\gamma$ occurring in t there exists a projecting rule in R over $g(\bar{z})\gamma$ leading to $z_i\gamma$,
- $s' \to^*_{R_{gemb}} t$ is either projection-closed with respect to R or empty.

These rule can be combined into a subset of the graph-embedded TRS, called contracting, for which the knowledge problems are decidable. However, the contracting definition also allows for a type of restricted permutation. For the results in this paper we don't need to include the permutation component of the contracting definition and by leaving it out we obtain a strict subset of contracting, introduced below, which is simpler and for which the knowledge problems are decidable.

Definition 8 (Permutation-free Contracting TRS).

A TRS R is said to be permutation-free contracting, $\approx$-free contracting for short, if for any non-subterm rule $l \to r$ in R, there exist a position p of l, a substitution σ

ranging over variables, and a projection-closed derivation $g \rightarrow^{+}_{R_{gemb}} d$ with respect to R such that $l|_p = g\sigma$ and $r = d\sigma$ is of depth 1.

Lemma 9 ([25, 10]).
Deduction and static equivalence are decidable for $\approx$-free contracting TRSs.

Proof.
Follows from [25] where it is shown that the knowledge problems are decidable for contracting TRS, and the fact that $\approx$-free contracting TRS are a strict subset of contracting. ◀

4 Decidable Theories

With the completion of the dichotomy results we next want to consider the question of non-trivial classes of theories for which all the above problems, knowledge, matching, and unification, decidable. One such example is the class of subterm convergent TRSs. These TRSs are very useful in the formal analysis of cryptographic protocols since they can model a number of cryptographic primitives. However, they require that the theory can be oriented into a restricted type of TRS. We introduce in this section another non-trivial example, the separate variable-permuting class of theories which are a restricted form of equational theory. We show that all of the above problems are decidable for this particular class of permutative theories.

4.1 Separate Variable-Permuting Theories

Let us begin by introducing the class of variable-permuting theories and then the class of separate variable-permuting (SVP) theories.

Definition 20 (Variable-Permuting Theory).
An axiom $l = r$ is said to be variable-permuting [21] if all the following conditions are satisfied:

1. the set of positions of l is identical to the set of positions of r,
2. for any non-variable position p of l, $l(p) = r(p)$,
3. for any $x \in Var(l) \cup Var(r)$, the number of positions of x in l is identical to the number of positions of x in r.

An equational theory E is said to be separate variable-permuting (SVP, for short) if for any $l = r \in E$ we have:

- $l = r$ is a variable-permuting axiom,
- l and r are rooted by the same function symbol that does not occur elsewhere in E.

An inner constructor symbol of E is a function symbol from E that never occurs at the root of any equality in E.

Example 21 (Intruder Theory with a Permutative Axiom).
Let us consider a theory used in practice to model a group messaging protocol [13]. For this protocol, the theory modeling the intruder can be defined [23] as a combination $R \cup K$ where

$$K = \{keyexch(x, pk(x'), y, pk(y')) = keyexch(x', pk(x), y', pk(y))\}$$

and

$$R = \left\{ \begin{array}{llll} adec(aenc(m, pk(sk)), sk) & \rightarrow & m & getmsg(sign(m, sk)) \rightarrow m \\ checksign(sign(m, sk), m, pk(sk)) & \rightarrow & ok & sdec(senc(m, k), k) \rightarrow m \end{array} \right\}.$$

R is a subterm convergent TRS and K is a SVP theory sharing with R the constructor symbol pk.

The class of SVP theories could be sufficient to specify intruder theories where one needs only a permutation of variables expressed via a single axiom, as in Example 21. Compared to a shallow theory like Commutativity, a SVP theory allows us to express a permutation involving terms with constructor symbols. Thus, SVP theories provide good examples for applying the combination methods developed in [16] to the deduction and static equivalence problems in the union of theories sharing only constructor symbols. We believe that SVP theories could be useful to approximate AC, and so to provide an alternative to AC in some cases. The AC theory is clearly very interesting in protocol analysis, for instance to specify the exclusive-or theory, but it is difficult to implement.

Contrary to AC, the decision procedures discussed in the following for the deduction and static equivalence problems in SVP theories are easy to implement in existing systems since they are very similar to the ones known in the class of subterm convergent theories, which is ubiquitous in protocol analysis.

4.2 Knowledge Problems in Separate Variable-Permuting Theories

The deduction problem is known to be decidable for the class of permutative theories [10], and so it is decidable for the class of SVP theories since SVP theories are permutative. The case of static equivalence is more complicated since it is known to be undecidable in permutative theories [15]. Therefore, we need to show that static equivalence becomes decidable when considering the particular case of SVP theories. Static equivalence has been shown to be decidable in shallow permutative theories [17]. However, the SVP theory K is not shallow. As shown below, the approach followed in [17] for the case of shallow permutative theories can be reused for the case of SVP theories. Actually, a decision procedure for static equivalence in any SVP theory E can obtained by computing a finite set of equalities bounded by $|E|$, where $|E| = \max_{\{l \mid l=r\in E\}} |l|$. To define this particular finite set of equalities, we first construct a frame saturation dedicated to SVP theories. Then, the recipes of the (deducible) terms included in that saturation are used to build an appropriate set of bounded equalities.

Let $St(t)$ be the set of subterms of a term t and let $St(\sigma)$ be the set of subterms of a substitution σ defined by $St(\sigma) = \bigcup_{x\in Dom(\sigma)} St(x\sigma)$.

Definition 22 (Frame Saturation for SVP Theories).

Let E be any SVP theory. For any frame $\phi = \nu\tilde{n}.\sigma$, let $ESt(\sigma) = St(\sigma) \cup \bigcup_{t\in ET} St(t)$ where ET is the set of terms $s[\vec{d}]$ such that $|s| \leq |E|$, $fn(s) \cap \tilde{n} = \emptyset$, $\vec{d} \in St(\sigma)$ and $s[\vec{d}] \leftrightarrow^{\epsilon}_{E} t$ for some term t.

The set of terms $sat_E(\phi)$ is the smallest set D such that:

(1)
$Ran(\sigma) \subseteq D$,

(2)
if $t_1, \ldots, t_n \in D$ and $f(t_1, \ldots, t_n) \in ESt(\sigma)$ then $f(t_1, \ldots, t_n) \in D$,

(3)
if $t \in D$, $t' \in ESt(\sigma)$, $t =_E t'$, then $t' \in D$,

For any frame $\phi = \nu\tilde{n}.\sigma$, σ_* is the substitution $\sigma\{\chi_u \mapsto u \mid u \in sat_E(\phi)\backslash Ran(\sigma)\}$ where χ_u is a fresh variable for any $u \in sat_E(\phi)\backslash Ran(\sigma)$, and ϕ_* denotes the frame $\nu\tilde{n}.\sigma_*$. Given a recipe ζ_u for each $u \in sat_E(\phi)\backslash Ran(\sigma)$, the substitution $\{\chi_u \mapsto \zeta_u \mid u \in sat_E(\phi)\backslash Ran(\sigma)\}$ is called a recipe substitution of ϕ and is denoted by ζ_ϕ. Let $Bt = \{t \mid |t| \leq |E|\}$. The set $Eq^B(\phi)$ is the set of equalities $t\zeta_\phi = t'\zeta_\phi$ such that $(t\zeta_\phi =_E t'\zeta_\phi)\phi$ and $t, t' \in Bt$.

The two following technical lemmas are similar to the ones developed in [17] for the case of shallow permutative theories.

Lemma 23.
Let E be any SVP theory. For any terms s and t satisfying the name restriction, if $s\phi_* =_E t\phi_*$ and $\psi \models Eq^B(\phi)$ then $(s\zeta_\phi)\psi =_E (t\zeta_\phi)\psi$.

Proof.
By induction on $\max(|s|, |t|, |E|) - |E|$.
■
Base case: if $\max(|s|, |t|, |E|) = |E|$, then it is true by definition of Eq^B.
■
Inductive step: consider s and t are terms such that $\max(|s|, |t|, |E|) > |E|$, $s = f(s_1, \ldots, s_n)$, $t = f(t_1, \ldots, t_n)$, and $s\phi_* =_E t\phi_*$. Then, there exist $f(l_1, \ldots, l_n) = f(r_1, \ldots, r_n) \in E$ and substitutions φ, μ such that
–
for $i = 1, \ldots, n$, $s_i\phi_* =_E l_i\varphi$ and $t_i\phi_* =_E r_i\varphi$. In all these matching problems, note that l_i and r_i are only built over inner constructors. Thus, these matching problems correspond to syntactic matching. For any variable x in $Var(l_i) = Var(r_i)$, only two cases can occur.
1.
$x\varphi$ is a term $u[w_1, \ldots, w_m]$ where $w_j \in Ran(\phi_*)$ for $j = 1, \ldots, m$. In that case, we define $x\mu = u[v_1, \ldots, v_m]$ where $v_j\phi_* = w_j$ for $j = 1, \ldots, m$.
2.
$x\varphi$ is a subterm $u|_p$ $(p \neq \epsilon)$ of a term $u \in Ran(\phi_*)$ such that any subterm $u|_q$ for $\epsilon \leq q < p$ is rooted by a inner constructor. Thus, $u|_p \in Ran(\phi_*)$ and so there exists a variable y such that $y\phi_* = u|_p$. In that case, we define $x\mu = y$.
With the above definition of μ, we have $\varphi = \mu\phi_*$.
–
Let $i = 1, \ldots, n$. By definition of μ, all the function symbol occurrences in $l_i\mu$ are already in s or in l_i. Thus, $|l_i\mu| \leq \max(|s_i|, |l_i|)$. Since $|s_i| < |s|$ and $|l_i| < |E|$, we have $|l_i\mu| < \max(|s|, |E|)$. Using the same argument, for $i = 1, \ldots, n$, $|r_i\mu| < \max(|t|, |E|)$.
Hence the induction hypothesis applies to $s_i\phi_* =_E (l_i\mu)\phi_*$ and $t_i\phi_* =_E (r_i\mu)\phi_*$, implying that
–
$(s_i\zeta_\phi)\psi =_E (l_i\mu\zeta_\phi)\psi$ for $i = 1, \ldots, n$,
–
$(t_i\zeta_\phi)\psi =_E (r_i\mu\zeta_\phi)\psi$ for $i = 1, \ldots, n$.
Thus we have:
$(s\zeta_\phi)\psi = f((s_1\zeta_\phi)\psi, \ldots, (s_n\zeta_\phi)\psi) =_E f((l_1\mu\zeta_\phi)\psi, \ldots, (l_n\mu\zeta_\phi)\psi) =_E f((r_1\mu\zeta_\phi)\psi, \ldots, (r_n\mu\zeta_\phi)\psi)$ $=_E f((t_1\zeta_\phi)\psi, \ldots, (t_n\zeta_\phi)\psi) = (t\zeta_\phi)\psi$ which completes the result.
◄

Lemma 24.
Let E be any SVP theory. For any frames ϕ and ψ, $\phi \approx_E \psi$ iff $\psi \models Eq^B(\phi)$ and $\phi \models Eq^B(\psi)$.

Proof.
(If direction) Let $Eq(\phi)$ be the set of all equalities $s\zeta_\phi = t\zeta_\phi$ such that $(s\zeta_\phi =_E t\zeta_\phi)\phi$.
Consider any $s\zeta_\phi = t\zeta_\phi \in Eq(\phi)$. By definition of ζ_ϕ and ϕ_*, we have $(s\zeta_\phi)\phi =_E s\phi_*$ and $(t\zeta_\phi)\phi =_E t\phi_*$. Since $(s\zeta_\phi)\phi =_E (t\zeta_\phi)\phi$, we obtain $s\phi_* =_E t\phi_*$. Then, by Lemma 23, we get $(s\zeta_\phi)\psi =_E (t\zeta_\phi)\psi$, which means that $\psi \models Eq(\phi)$. In a

symmetric way, we show that $\phi \models Eq(\psi)$. Then, we can conclude since $\phi \approx_E \psi$ iff $\psi \models Eq(\phi)$ and $\phi \models Eq(\psi)$. ◀

Proof.

Decidability of deduction follows from the decidability of deduction in permutative theories [10], *SVP* theories being permutative. Static equivalence is decidable thanks to Lemma 24. ◀

The combination method developed in [16] can be directly applied to a union $R \cup E$ where R is a convergent TRS with decidable deduction and static equivalence, and E is a *SVP* theory. In that case, the function symbols shared by R and E must be constructors for R and inner constructors for E. Notice that with a shallow permutative theory E, constructor symbols can only occur in the ground terms appearing in E, and this is clearly a limitation to the application of the combination method. Thus, considering *SVP* theories instead of shallow ones allows us to get more significant applications.

Corollary 26.

Let R be any convergent TRS where both deduction and static equivalence are decidable, and E any *SVP* theory such that the function symbols shared by R and E are constructors for R and inner constructors for E. Then, both deduction and static equivalence are decidable in $R \cup E$.

4.3 Unification and Matching in Separate Variable-Permuting Theories

Finally, let us consider the unification and matching problems in *SVP* theories. Similar to the static equivalence case, we cannot rely on the decidability of unification in the whole class of permutative theories, since unification have been shown to be undecidable in permutative theories [26] and even in variable permutative theories [15]. Fortunately, *SVP* theories are examples of theories with the property of being closed by paramodulation, a notion formally introduced in [24, 20, 18]. Actually, the closure by paramodulation follows from the absence of non-variable overlap between two axioms of any *SVP* theory. Since unification is decidable and finitary in theories closed by paramodulation [24, 20], we have that unification is finitary in *SVP* theories. Since *SVP* theories are regular and collapse-free, any unification algorithm in *SVP* theories leads to a unification algorithm with free symbols, thanks to the combination method known for unification algorithms in regular collapse-free theories [27], the empty (aka free) theory being regular collapse-free. Consequently, unification with free constants, and so matching is also finitary in *SVP* theories.

Theorem 27.

Unification and matching are decidable in any *SVP* theory.

Finally, we can put all the above results together to obtain the following.

Theorem 28.

Unification, matching, deduction, and static equivalence are decidable for any *SVP* theory.

Theorem 25.

Deduction and static equivalence are decidable in any *SVP* theory.

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